

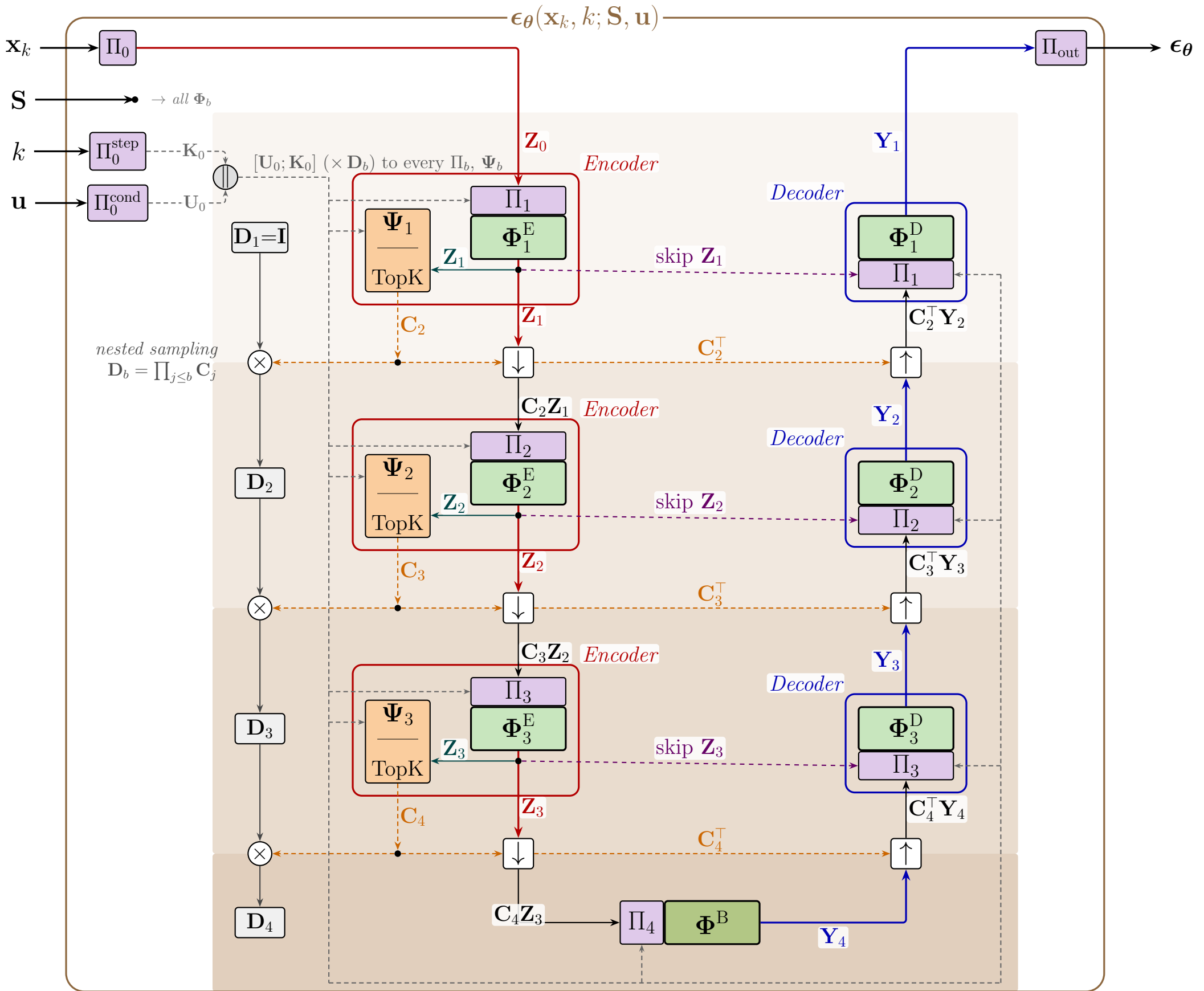


Fig. 1. U-GNN denoiser ($B=4$ depths, incl. bottleneck) as a vertical encoder–decoder U. Each block stacks its per-block projection Π_b (fuse with $\mathbf{D}_b[\mathbf{U}_0; \mathbf{K}_0]$) on the GNN Φ_b . Selectors Ψ_b produce $\mathbf{C}_{b+1}$ (reused transposed by the matched decoder up-sampling); the left pipe nests $\mathbf{D}_b = \prod_{j \leq b} \mathbf{C}_j$. The graph shift operator $\mathbf{S}$ enters once and parametrises every GNN block. The module frame ϵ_θ leaves the raw I/O ($k, \mathbf{u}, \mathbf{x}_k, \mathbf{S}, \epsilon_\theta$) outside. The **encoder** and **decoder** blocks (outlined) are detailed in Fig. 2.

